

Pituitary and Hypothalamus

Instructor: Christopher A. Haines, MD,
(christopher.haines@jefferson.edu)

Block: Endo/Uro/Repro

Thread: Physiology

Conflict of Interest: Dr. Haines has no conflicts of interest to disclose.

Learning Objectives

By exam time, you should be able to do the following:

1. Describe the major structures of the hypothalamic-pituitary axis.
2. Describe the hormones produced in the anterior and posterior pituitary and targets of each.
3. Describe the production, secretion, regulation, and functions of growth hormone and IGF-1 within the GH/IGF-1 axis, including the insulin-like parallels of IGF-1.
4. Identify the insulin-like functions and the anti-insulin functions of growth hormone.
5. Define dwarfism, gigantism, and acromegaly.
6. Identify the signs and symptoms of acromegaly.

Study Questions

1. What are the major structures of the hypothalamic-pituitary axis?
2. What type of intercellular communication occurs at the posterior lobe of the pituitary? What are the major hormones released by the posterior pituitary?
3. What type of intercellular communication occurs between the hypothalamus and the anterior pituitary? What type of intercellular communication occurs at the level of the anterior pituitary cells?
4. What are the major hormones released by the anterior pituitary? What feedback mechanisms influence release of hormones of the anterior pituitary?
5. What does growth hormone regulate? What is the structure of growth hormone? How is growth hormone synthesis and release influenced by the hypothalamus?

6. Where are growth hormone releasing hormone receptors located? How does GHRH influence somatotropes in the anterior pituitary? How does somatostatin influence somatotropes?
7. How does growth hormone secretion vary during the day and throughout the life cycle? What are some major mediators of growth hormone secretion?
8. What is the structure of the growth hormone receptor? Where are growth hormone receptors located?
9. What are insulin-like growth factors? Where are IGFs synthesized? What are the effects of IGFs on various tissues and systems?
10. What are the major characteristics of growth hormone excess? Deficiency?

A. Hypothalamic-Pituitary Axis

1. The pituitary is located posterior to the bridge of the nose at the base of the brain just below the hypothalamus. The hypothalamus controls most pituitary functions. The anterior and posterior lobes of the pituitary are connected to hypothalamus by the pituitary stalk (Figure 1). Hormones secreted by the pituitary gland are listed in Table 1.
2. **Posterior Pituitary:** Neural input from other areas of the brain regulates much of the hypothalamic activity. When the cell bodies of the supraoptic and paraventricular nuclei of the hypothalamus are induced to synthesize ADH and oxytocin, these hormones are transported down the axons to terminals proximal to the capillaries in the posterior pituitary lobe.

Activation of action potentials releases the hormones for uptake by the capillaries, and movement into the systemic circulation.

3. **Anterior Pituitary:** Other nuclei within the hypothalamus synthesize releasing/inhibiting hormones (short peptides or amino acids) that are released from the nerve terminals at the median eminence and taken up by arterially fed capillaries. These capillaries feed into the long portal veins which travel through the pituitary stalk to the capillaries in the anterior lobe.
 - a. The releasing/inhibiting hormones regulate synthesis and secretion of hormones in the anterior pituitary. The hormones secreted by the anterior pituitary are then delivered to the systemic venous circulation.
 - b. Short portal veins connect the posterior pituitary lobe to the anterior pituitary lobe, perhaps moving components from the posterior pituitary to regulate the anterior pituitary.

Figure 1

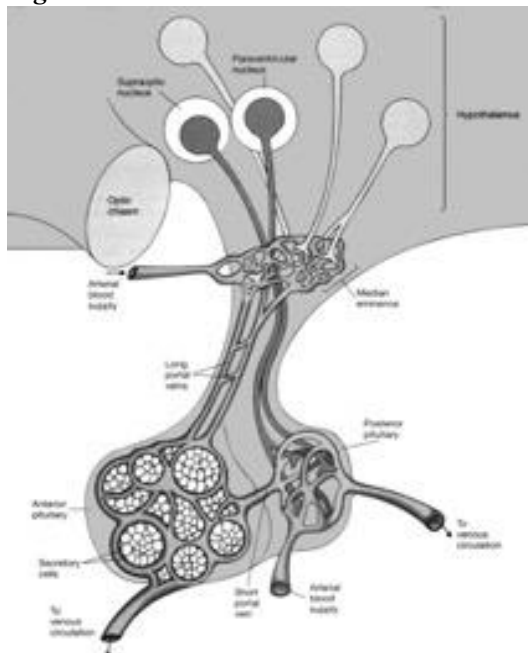


TABLE 1

Anterior pituitary lobe cells	Hormone secreted	Regulated by (↑) stimulation (↓) inhibition	Hypothalamic hormone
corticotropes releasing	adrenocorticotropin (ACTH)	(↑)	corticotropin-
thyrotropes	thyroid-stimulating hormone (TSH)	(↑)	hormone (CRH) thyrotropin releasing hormone (TRH)
gonadotropes releasing	luteinizing hormone (LH)	(↓) (↑)	somatostatin gonadotropin-
somatotropes	follicle-stimulating hormone (FSH) growth hormone (GH)	(↑)	hormone (GnRH) growth hormone-releasing hormone (GHRH)
lactotropes	prolactin (PRL)	(↓) (↑) (↓)	somatostatin prolactin-releasing factor (PRF) prolactin-inhibitory hormone (PIH) (↓) somatostatin

Neuron bodies in hypothalamus	Hormone synthesized	Released in posterior pituitary lobe
supraoptic nuclei paraventricular nuclei	antidiuretic hormone (ADH, vasopressin) oxytocin	by action potentials by action potentials

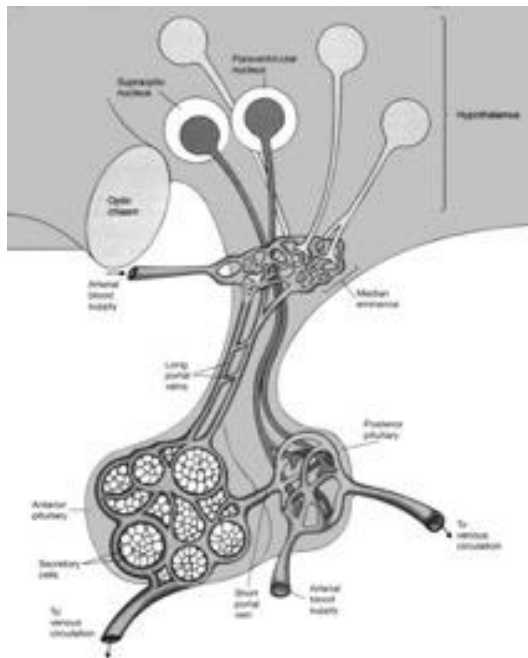
B. Growth Hormone (GH)

1. Regulates longitudinal bone growth and the metabolism of glucose, amino acids and fatty acids.

C. GH synthesis

1. GH is a single polypeptide protein with 2 disulfide bonds, synthesized by the somatotrope cells of the anterior pituitary.
2. GH gene transcription is increased by growth hormone-releasing hormone (GHRH) from the hypothalamus.
3. The prehormone signal sequence is removed before storage of GH in dense granules.

Figure 2



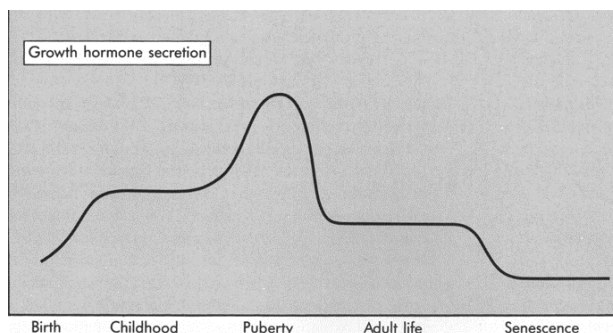
D. GH secretion

1. Basal GH level in adults is <3 ng/ml.
2. GH secretion is stimulated by the hypothalamic hormone **GHRH (Figure 2)**. GHRH is released into the circulatory system at the median eminence, travels through the long portal veins and circulates through the capillary bed in the anterior pituitary where the somatotrophic cells are located.
3. **GHRH receptors** (GPCRs) are expressed almost exclusively on the somatotropes in the anterior pituitary. GHRH receptor activation results in increased cAMP synthesis and release of intracellular Ca^{2+} by the somatotropes.
4. GH secretion is inhibited by **somatostatin** which blocks GHRH signaling, resulting in decreased cAMP and intracellular Ca^{2+} concentrations in the somatotropes. Somatostatin is made in the hypothalamus and secreted in response to high GH and IGF levels.
5. GH secretion is **pulsatile**. The pulsed release of GHRH causes a shift in the GHRH:somatostatin ratios in the pituitary portal circulation. The periodic absence of GH is important for maintaining the synthetic functions of GH while minimizing its insulin-antagonistic actions (enhanced lipolysis, impaired glucose uptake and utilization).
6. GH secretion exhibits **large amplitude oscillations**. Plasma GH concentration increases dramatically during deep sleep (slow-wave) preceded by a GHRH peak. Daytime naps elicit pulses of GH secretion, more being released during afternoon than morning. Deprivation of slow-wave sleep decreases GH secretion.

E. Mediators of growth hormone secretion

1. Stimulators: Hypoglycemia, slow wave sleep, puberty (Figure 5), stress, exercise
2. Inhibitors: Glucose, REM sleep

Figure 5



F. GH metabolism

1. GH is metabolized by the liver with a $t_{1/2} \sim 6-20$ min.

G. GH Receptors

1. The GH receptor is synthesized as a single chain glycoprotein. The receptors dimerize by two receptors binding GH at different sites (**Figure 6**). Receptor dimerization initiates a phosphorylation cascade that leads to new protein synthesis. Excess GH prevents dimerization.
2. GH receptor synthesis is stimulated when the circulating GH concentration is low. Insulin and estrogen also promote GH receptor synthesis. GH receptor synthesis is inhibited by high circulating concentrations of GH and by fasting.

Figure 6

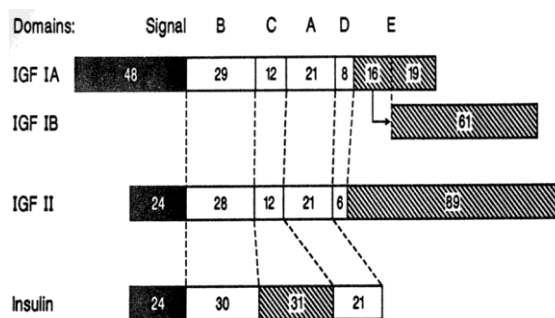
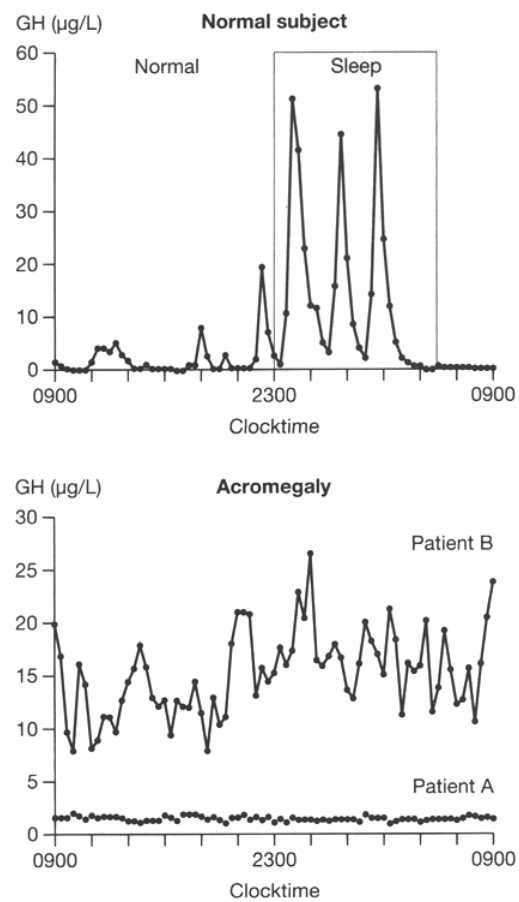


Figure 7



H. Insulin-like growth factors (IGFs or somatomedins)

1. The GH growth-promoting effects are observed 12-18 hrs. after GH levels increase and are due to induction of IGF synthesis in the liver.
2. IGF-I and IGF-II resemble proinsulins (**Figure 7**). The A and B domains are each 50% homologous, while the C domains are unique.

I. IGF Receptors

1. IGF-I receptor is a tetramer that resembles the insulin receptor, a tyrosine kinase.
2. IGF-II Receptor is a monomer. It binds IGF-I poorly and doesn't bind insulin.

J. IGF Binding Proteins

1. 75% of IGF-I and IGF-II are complexed to IGF binding proteins. The binding proteins (4-6 variants) protect IGFs from degradation, supporting their long-term actions (potentiation of DNA synthesis, etc.). The $t_{1/2}$ of the IGFs is approximately 12 hours.

K. GH and IGF Actions

1. GH
 - a. Liver: increase gluconeogenesis, protein synthesis
increase IGF production
 - b. Adipose tissue: increase lipolysis
decrease adiposity
decrease glucose uptake
 - c. Muscle: increase protein synthesis
increase muscle mass
decrease glucose uptake
-

2. IGF-I

- a. Bone: increase protein, collagen synthesis
increase linear growth
increase cell proliferation

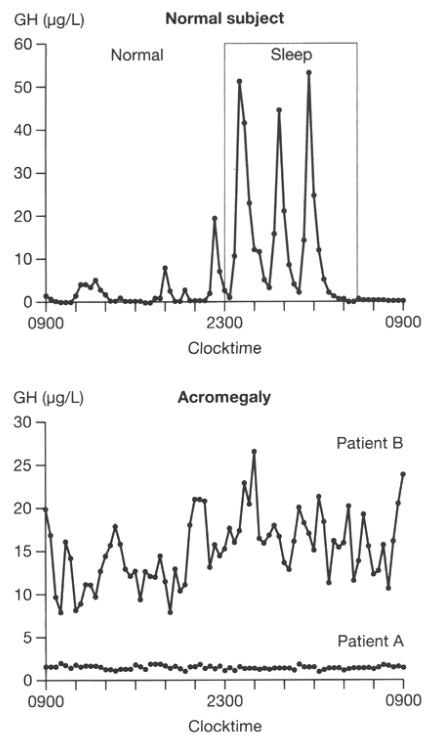
3. IGF-II

- a. Organs increase protein synthesis
- b. increase organ size
increase cell proliferation
(heart, lung, kidney, pancreas, intestine, parathyroid, skin, connective tissue)

L. GH Pathophysiology

- 1. Gigantism/Acromegaly: excess GH secretion (**Figure 9**)

Figure 9 – Growth hormone levels in normal subject and in acromegaly



- a. gigantism
 - i. onset during childhood or puberty
 - b. acromegaly
 - i. onset after puberty
 - c. impaired function
 - i. pituitary tumor
 - d. clinical symptoms:
 - i. accelerated growth
 - ii. insulin resistance (GH-induced decrease in insulin receptor number?)
 - iii. mild carbohydrate intolerance
 - e. treatment
 - i. tumor removal
 - ii. somatostatin analogs
 - iii. dopaminergic agents
 - iv. GH antagonist
2. Dwarfism: GH deficiency
- a. impaired function
 - i. at the hypothalamus
 - 1. no GHRH synthesis
 - ii. at the pituitary
 - 1. no GH synthesis or nonfunctional GH
 - iii. at the liver
 - 1. no IGF-I and IGF-II synthesis
 - iv. target tissue receptor defects: numbers, binding, signaling
 - c. clinical symptoms
 - v. short stature
 - vi. delayed bone maturation
 - vii. mild obesity
 - viii. delayed puberty
 - d. treatment
 - i. hormone replacement therapy

M. Case History:

A 48-year-old man complaining of impotence sought medical attention.

Questioning revealed: shaving less frequently
shoe size had increased from 9C to 11EEE
over 5 yr
dental plate altered 3x in 6 yr
friends had noted changes in his appearance
tingling in fingers; joint pain

Physical exam: coarse facial features, bulbous nose
tongue enlarged
teeth widely spaced
loss of both lateral visual fields
hands and feet enlarged, fingers spadelike
liver enlarged

fasting glucose: 150 mg/dl

fasting GH: 40 ng/ml; unaltered by oral glucose load

MRI: large pituitary mass protruding upward from
sella turcica

Diagnosis: Acromegaly

Answers to Study Questions

1. What are the major structures of the hypothalamic-pituitary axis?

The major structures include the hypothalamus, the trophic cells of the anterior pituitary, and the various end organs. End organs of note include the liver (producing insulin-like growth factor), the thyroid, the adrenal cortex, the ovaries, and the testes.

2. What type of intercellular communication occurs at the posterior lobe of the pituitary? What are the major hormones released by the posterior pituitary?

The posterior pituitary releases peptide hormones in a neuroendocrine fashion. The major hormones released by the posterior pituitary are anti-diuretic hormone (ADH; also known as arginine vasopressin) and oxytocin.

3. What type of intercellular communication occurs between the hypothalamus and the anterior pituitary? What type of intercellular communication occurs at the level of the anterior pituitary cells?

Peptide hormones from the hypothalamus are released at the median eminence in a neuroendocrine fashion. These hormones signal the trophic cells of the anterior pituitary to synthesize and release peptide hormones into the general circulation in an endocrine fashion.

4. What are the major hormones released by the anterior pituitary? What feedback mechanisms influence release of hormones of the anterior pituitary?

Major hormones released by the anterior pituitary include growth hormone, thyroid stimulating hormone, adrenocorticotropin, follicle-stimulating hormone, luteinizing hormone, and prolactin. Negative feedback mechanisms regulate release of anterior pituitary hormones, with anterior pituitary hormones themselves as well as end-organ hormone (like IGF-1, thyroid hormone, cortisol, etc.) also feeding back on the anterior pituitary in a negative fashion.

5. What does growth hormone regulate? What is the structure of growth hormone? How is growth hormone synthesis and release influenced by the hypothalamus?

Growth hormone increases gluconeogenesis and protein synthesis in the liver, increases lipolysis and decreases glucose uptake in adipose tissue, and increases protein synthesis and decreases glucose uptake in muscle. In addition, growth hormone increases somatomedin (IGF-I and IGF-II) synthesis and release.

GH is a single polypeptide protein with 2 disulfide bonds, synthesized by the somatotrope cells of the anterior pituitary.

GH secretion is stimulated by the hypothalamic hormone GHRH. GH secretion is inhibited by somatostatin. Somatostatin is made in the hypothalamus and secreted in response to high GH and IGF levels.

6. Where are growth hormone releasing hormone receptors located? How does GHRH influence somatotropes in the anterior pituitary? How does somatostatin influence somatotropes?

Extracellular receptors to GHRH are located on the somatotropes of the anterior pituitary. GHRH causes somatotropes to increase synthesis and release of GH. Somatostatin inhibits synthesis and release of GH.

7. How does growth hormone secretion vary during the day and throughout the life cycle? What are some major mediators of growth hormone secretion?

GH secretion is pulsatile, and also exhibits large amplitude oscillations. Plasma GH concentration increases dramatically during deep sleep (slow-wave). Deprivation of slow-wave sleep decreases GH secretion.

GH secretion varies during the life cycle and is highest during puberty.

8. What is the structure of the growth hormone receptor? Where are growth hormone receptors located?

The GH receptor is a single chain glycoprotein extracellular receptor. The receptors dimerize by two receptors binding GH at different sites.

GH receptor synthesis is stimulated when the circulating GH concentration is low. Insulin and estrogen also promote GH receptor synthesis. GH receptor synthesis is inhibited by high circulating concentrations of GH and by fasting.

GH receptors are ubiquitous and growth hormone acts on many tissues.

9. What are insulin-like growth factors? Where are IGFs synthesized? What are the effects of IGFs on various tissues and systems?

Insulin-like growth factors are peptide hormones synthesized in the liver. IGF-I has its main effect on bone. It increases protein and collagen synthesis as well as increases cell proliferation. Overall, IGF-I increases linear growth of bone.

10. What are the major characteristics of growth hormone excess? Deficiency?

Growth hormone excess in childhood or puberty leads to gigantism. Growth hormone excess after puberty leads to acromegaly. Growth hormone deficiency leads to dwarfism, with short stature, delayed bone maturation, mild obesity, and delayed puberty.